Volume of Revolution: Parametric Equations

Questions 1–13

Question 1. For each of the following pairs of parametric equations, the region is bounded by the given curve, the x-axis, the y-axis and any line if given. The region is rotated around the x-axis until a solid of revolution is generated. In each case, determine the volume of the solid generated giving your answer in exact form:

a) $x = t^2$, $y = t^3$ for $t \geq 0$ with line $x = 4$
b) $x = 2t^3$, $y = t^2$ for $t \geq 0$ with line $x = 16$
c) $x = t^3 + 1$, $y = t^2$ for $t \geq 0$ with line $x = 28$
d) $x = \sin t$, $y = \sqrt{\cos t}$ for $0 \leq t \leq \frac{\pi}{2}$

Question 2. For each of the following pairs of parametric equations, the region is bounded by the given curve, the x-axis, the y-axis and any line if given. The region is rotated around the y-axis until a solid of revolution is generated. In each case, determine the volume of the solid generated giving your answer in exact form:

a) $x = \sqrt{1 - t^2}$, $y = 1 + t^2$ for $0 \leq t \leq 1$
b) $x = 2\cos t$, $y = \tan t$ for $0 \leq t \leq \frac{\pi}{2}$ with line $y = 1$
c) $x = \sin^2 t\sqrt{\cos t}$, $y = -\cos t$ for $0 \leq t \leq \frac{\pi}{2}$

Question 3. A curve is defined by parametric equations

$$x = e^{\frac{1}{2}t} \quad \text{and} \quad y = \sqrt{t-2}.$$

A region R is enclosed by the curve, the line $x = e^2$ and $x = e^3$. Show that the volume of the solid generated when R is rotated about the x-axis by 2π radians is given by $2\pi e^3$.

Question 4. A curve is defined by parametric equations

$$x = \sqrt{\ln|t+1|} \quad \text{and} \quad y = t^2.$$

A region R is enclosed by the curve, the line $y = 1$ and $y = 9$. Show that the volume of the solid generated when R is rotated about the y-axis by 2π radians is given by $2\pi(8\ln 2 - 1)$.

Question 5. A funnel may be modelled by a curve C defined parametrically by

$$x = \frac{4}{t} \quad \text{and} \quad y = \ln t$$

for $t \geq 1$. The region bounded by the curve C, the y-axis, and the lines $y = 0$ and $y = \ln 4$ is rotated through 2π radians about the y-axis to form a solid. All lengths are measured in centimetres.

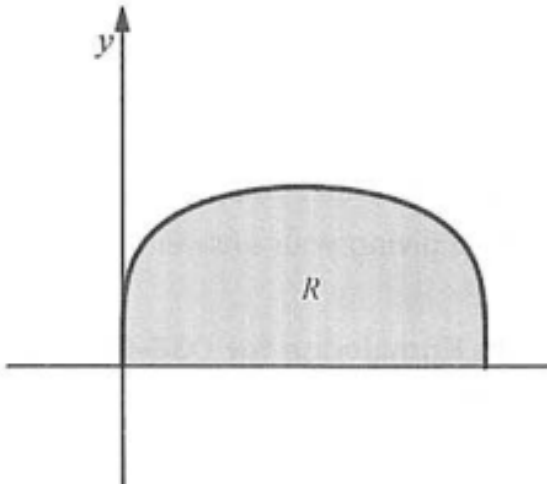
a) Find the exact volume of the funnel.

b) State a limitation of the model.

Question 6. A curve is defined by parametric equations

$$x = 1 - \cos t \quad \text{and} \quad y = \sqrt{\sin t}$$

for $0 \leq t \leq k$ for constant k and is shown in the diagram below. A region R is enclosed by the curve and the x-axis.



a) State the value of k.

b) Determine the volume of the solid generated when R is rotated about the x-axis by 2π radians.

Question 7. A curve is defined by parametric equations

$$x = \sqrt{t} \quad \text{and} \quad y = e^t$$

for $0 \leq t \leq k$ for constant k. Given that the volume of the solid generated when R is rotated about the y-axis by 2π radians is $\pi(3e^4 + 1)$, determine the value of k.

Question 8.

a) By using the substitution $u = \sin\theta$, show that

$$\int \cos^3\theta \, d\theta = \sin\theta - \frac{1}{3}\sin^3\theta + c.$$

b) Find the volume formed by the region enclosed by the curve defined by $x = \sin\theta$ and $y = \cos\theta$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ when rotated 2π radians across the x-axis.

Prerequisite Knowledge for Question 9: Calculus of Hyperbolic Functions and Improper Integrals

Question 9. A curve C is defined by parametric equations

$$x = \cosh t - \sinh t \quad \text{and} \quad y = \cosh t + \sinh t$$

for $t \leq 0$. An unbounded region is formed when the region between the curve C, the x-axis and the line corresponding to $t = 0$ is rotated through 2π about the x-axis.

a) Show that the curve can be expressed as $y = \dfrac{1}{x}$ for $x \geq 1$.

b) Hence, determine the exact volume of the solid formed.

Question 10. The inside surface of a nozzle is modelled by the parametric curve C

$$x = ae^{-t} \quad \text{and} \quad y = bt^2$$

for $0 \leq t \leq 2$ where a and b are positive constants. The curve is rotated 2π radians about the y-axis. The bottom of the nozzle sits on the x-axis. All lengths are measured in centimetres. It is required that the nozzle has diameter 4 cm and a height of 3 cm.

a) Determine the values of a and b according to the model.

b) Find the volume of the nozzle giving your answer to three significant figures.

c) State a limitation of the model.

Question 11. A glass ornament is made by rotating a smooth curve about a vertical axis. The profile of a glass ornament is modelled by a curve C defined parametrically by

$$x = \sqrt{t^2 - 1} \quad \text{and} \quad y = t + \ln t$$

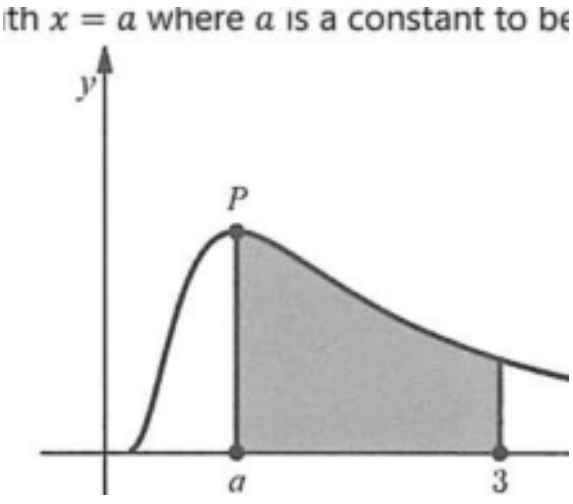
for $1 \leq t \leq 3$. All lengths are measured in centimetres. The region between the curve C and the y-axis is rotated through 2π radians about the y-axis to form a solid.

a) Find the maximum diameter of the ornament, giving your answer to the nearest mm.
b) Find the maximum height of the ornament, giving your answer to the nearest mm.
c) Find the volume of the glass ornament to three significant figures.
d) The ornament is made from glass with density 2.5 g cm^{-3}. Find the mass of the ornament, giving your answer to two significant figures.

Question 12. A curve is defined by parametric equations

$$x = \frac{2}{t} \quad \text{and} \quad y = \frac{t^2}{e^t}$$

for $t > 0$. The maximum of the curve is at the point P with $x = a$ where a is a constant to be found.



A region R is enclosed by the curve, the line $x = a$ and $x = 3$.

a) Find the coordinates of P.
b) Show that the volume of the solid generated when R is rotated about the x-axis by 2π radians is given by

$$\pi\left(\frac{29}{18e^{4/3}} - \frac{13}{2e^4}\right).$$

Question 13. A curve is defined by parametric equations

$$x = \ln(t-3) \quad \text{and} \quad y = \frac{8}{\sqrt{t}}$$

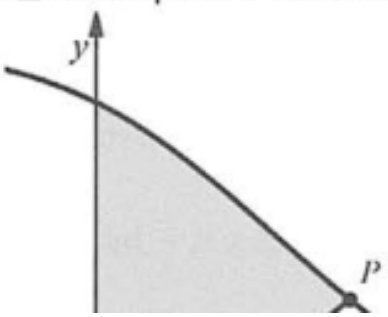
for $t \geq 3$. The point P has x-coordinate of $\ln 13$.

$$\pi\left(\frac{\quad}{18e^{\frac{4}{3}}} - \frac{\quad}{2e^4}\right)$$

curve is defined by parametric equat

$$x = \ln(t-3) \text{ and } y = \frac{8}{\sqrt{t}}$$

or $t \geq 3$. The point P has x-coordinate



The shaded region is enclosed by the curve, the y-axis and the line which passes through the point P and the origin. Determine the exact volume of the solid generated when the shaded region is rotated about the x-axis by 2π radians.